WetlandSentinel: An Ellipsoidal Amphibious Robot for Wetland Ecological Monitoring

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***Abstract*—Wetland monitoring requires water-quality measurements at the water surface, and access to monitoring locations may require movement through both shallow water and terrestrial terrain. This paper presents WetlandSentinel, a remotely operated ellipsoidal amphibious robot whose internal motor transmits torque through a gear mechanism to drive rolling locomotion on land. The robot integrates movable-mass steering, LoRa telemetry, and multimodal sensing. The pH and turbidity sensors measure local water conditions, and the onboard K210 vision module detects objects in the surrounding environment. In an open-area field test, LoRa remote control was maintained as the robot moved away from the operator to a separation of approximately 600 m. Separate cup-based bench validation of the water-quality module produced a pH mean absolute error of 0.025 and a maximum turbidity deviation of 1.00%. Controlled tank trials evaluated traversal through simulated stone and vegetation obstacles; over a 3 m path, the maximum normalized lateral deviation was 2.6%. At the 15 cm tested working depth, a separate 3 h enclosure test showed no visible internal water ingress. Together, these tests support a mobile platform for acquiring multimodal wetland observations that could serve as inputs to future spatiotemporal environmental data mining.**

***Index Terms*—wetland monitoring, ellipsoidal robot, LoRa, multimodal sensing, water quality**

I. Introduction

Wetlands regulate water quality, support biodiversity, and store carbon, but their condition continues to decline globally [1]. Monitoring requires observations across locations with different access conditions. Unmanned aircraft systems (UAS) provide environmental imagery [2], while wireless sensor networks support distributed in situ measurements [3]. Mobile unmanned surface vehicles (USVs) combine water-quality sensing with remote operation [4], including LoRa telemetry [5]. Access across shallow water and terrestrial terrain motivates an amphibious sensing platform.

WetlandSentinel uses an ellipsoidal body with internal actuation and replaceable end modules (Fig. 1). Water-quality probes contact the surrounding water at one end, while a camera operates behind a transparent cover at the other. LoRa links the platform to a handheld terminal for remote commands and sensor feedback.

The project makes five contributions to multimodal wetland data acquisition: (1) a sealed ellipsoidal architecture for amphibious mobility; (2) an internal gear transmission and sliding counterweight for propulsion and steering; (3) dual-axis fuzzy gain-scheduled PID control; (4) LoRa remote operation tested at approximately 600 m; and (5) modular water-quality and vision interfaces that allow sensing functions to be extended without rebuilding the central drive.

II. Monitoring Requirements and Design Rationale

Five requirements guided the prototype. It should move through shallow water, wet soil, and scattered vegetation while protecting the central electronics from water ingress. It should obtain direct in situ water-quality measurements near the surface rather than infer them only from imagery. The operator should receive sensor and attitude information through remote telemetry without retrieving the robot. Replaceable probes and end modules should allow sensing functions to change without redesigning the central drive and steering systems. Finally, the controller should maintain stable motion and responsive steering as the terrain and obstacles change.

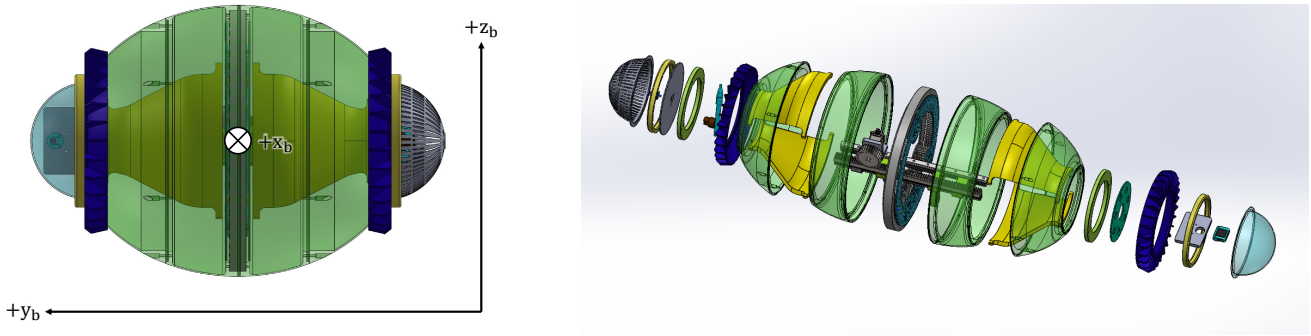
Enclosed rolling robots employ diverse internal actuation mechanisms [6], and amphibious spherical robots demonstrate terrestrial and aquatic operation [7]. WetlandSentinel uses an ellipsoid to provide a defined major axis for sensor placement and sliding-mass steering. One end can be directed toward the water for pH and turbidity measurements, while the opposite end houses the camera behind a transparent cover. Its continuous exterior reduces protruding components and avoids an external steering rudder or drivetrain.

Existing approaches offer different access capabilities: screw-driven robots have been developed for soft ground and dense wetland plants [8], and USVs combine water-quality monitoring with surface cleaning [9]. WetlandSentinel emphasizes an enclosed mechanism and movement between land and water. Compared with fixed nodes, it can relocate its measurement position; compared with aerial imaging, its immersed probes measure water quality directly. This flexibility introduces the mechanical and control demands of sealing and active balancing.

III. System Design

A. Mechanical Architecture

The robot has a four-part split ellipsoidal shell fabricated by stereolithography. Grooved interfaces, pressure rings, and silicone seals isolate electronics while allowing module access. Its smooth profile was selected for shallow water, wet soil,



(a) Body-fixed coordinate frame.

(b) Exploded internal architecture.

Fig. 1. WetlandSentinel body-fixed frame and internal architecture.

and low vegetation. The prototype was tested at a working water depth of approximately 15 cm. Table I summarizes its dimensions, mass, and implemented configuration. The nominal 3.2 h endurance is a bench estimate based on the installed battery and measured operating load.

The locomotion mechanism consists of a rotating ellipsoidal shell and an internal frame. An MG540 DC geared motor mounted on the internal frame drives the shell through a three-stage gear train engaging an internal ring gear, with a nominal output-to-input speed ratio of 1:10. The motor therefore drives the outer shell relative to the internal frame. During steady motion, the internal frame remains approximately upright while translating and turning with the robot, with small attitude changes during acceleration and steering.

A body-fixed frame is defined with x_b along the nominal forward direction, y_b along the ellipsoid major axis from the sensing end toward the camera end, and z_b upward. The shell rotates primarily about y_b. On land, this rotation produces rolling locomotion through shell–ground contact; in water, the peripheral blades rotate with the shell and interact with the surrounding water. Propulsion and steering are actuated by the shell-drive motor and sliding counterweight, respectively.

Blade geometry was compared in tank trials over a nominal 2 m test distance, using two video-derived trajectories per configuration. The analysis used a common forward-displacement interval of 0.1–1.5 m from each record's first point. Coordinates were reset at the interval start, and lateral displacement was measured from a line parallel to the recorded forward axis. The mean of the two maximum absolute lateral displacements was 64.7, 11.7, and 26.6 cm for the smooth shell, 10 mm blades, and 20 mm blades, respectively (Fig. 2). The smaller mean lateral displacement supported selecting the 10 mm configuration.

Directional control is provided by a stepper-driven counterweight that translates along y_b. Moving the mass away from the center shifts the center of mass along the major axis and generates a gravitational roll moment about x_b. If the counterweight mass is m_c, its displacement from the center is y_c, and the roll angle about x_b is ϕ, the magnitude of this moment can be approximated by

$$|\tau_{x_b}| = m_c g |y_c \cos \phi|. \tag{1}$$

The resulting roll bias lowers one end of the ellipsoid while raising the other. This motion is used both to orient the water-quality probes toward the water and to provide a steering bias through asymmetric contact or hydrodynamic interaction. Steering is therefore an indirect consequence of controlled roll rather than a directly actuated yaw motion.

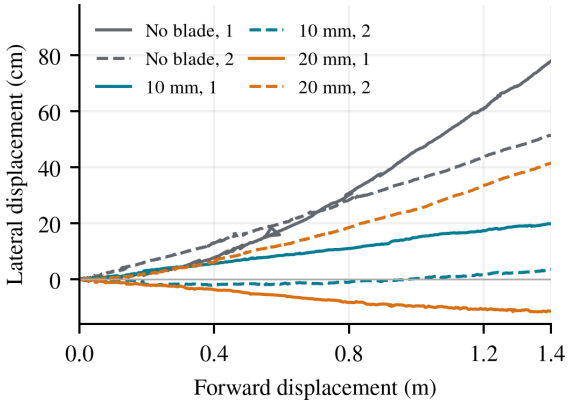
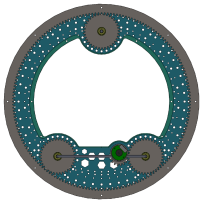


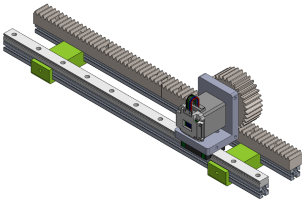
Fig. 2. Blade-comparison trajectories over the common analysis interval (two records per configuration).

TABLE I
WetlandSentinel Prototype Specifications

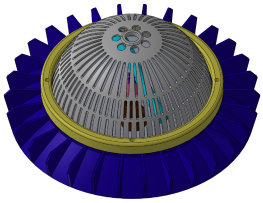
Item	Specification
Dimensions	420 mm × 420 mm × 640 mm
Mass	approximately 13 kg
Drive	MG540 DC geared motor
Transmission	Three-stage; output/input speed ratio 1:10
Working water depth	approximately 15 cm
Tested remote control	approximately 600 m
Estimated endurance	approximately 3.2 h (bench estimate)
Balance/steering	35-mm stepper; rack and linear guide
Attitude sensing	MPU6050 IMU
Water-quality sensing	SEN0161 pH; TS-300B turbidity
Vision	Sentry2 K210 module
Control/telemetry	ESP32 controllers; SX1278 LoRa



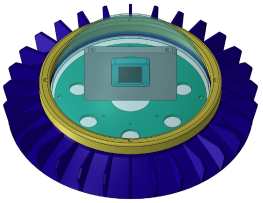
(a) Gear-drive module engaging the internal ring gear.



(b) Stepper-driven sliding-balancing mechanism.



(c) pH/turbidity sensing end module.



(d) K210 vision end module behind the acrylic dome.

Fig. 3. CAD views of the principal mechanical and sensing modules: shell drive, sliding balance, water-quality sensing, and vision.

B. Module Integration and Maintainability

The two circular end interfaces carry the sensing and camera assemblies, while the central split shell contains the drive, battery, controllers, and translating mass (Fig. 3). Turntable bearings decouple the rotating blade rings from the protected end modules. Pressure rings distribute the sealing load around each aperture. An acrylic dome preserves the camera field of view, whereas a mesh PETG dome allows water contact while shielding the probes.

The end modules allow sensor replacement and seal inspection without separating the central shell, while drivetrain service leaves the sensor assemblies in place. In a separate enclosure test, the sealed central enclosure remained at the tested working depth of 15 cm in tap water for 3 h; post-test disassembly found no visible internal water ingress, and the electronics remained operational.

C. Electronics and Communication

Three ESP32-based subsystems handle motion control, environmental sensing, and vision. The motion controller reads the MPU6050 inertial measurement unit (IMU), interprets remote commands, and drives the DC and stepper motors. The sensing controller acquires pH and turbidity measurements for telemetry, while the vision controller interfaces with the K210, which performs inference locally.

The handheld terminal uses dual joysticks and an OLED for operation and feedback, with an ESP32 and SX1278 transceiver providing the LoRa link. Motion commands are processed upon receipt, whereas water-quality reports are generated every 5 s. The link supports above-water or near-surface operation.

D. Edge Processing and Control

The sensing controller converts analog probe voltages into reported values. For pH, a rolling buffer holds 40 ADC samples; the minimum and maximum are discarded, and the remaining 38 values are averaged before conversion.

Accelerometer and gyroscope observations are fused with a Kalman filter [10]. Figure 4 summarizes the feedback paths. The pitch and roll loops use fuzzy gain-scheduled PID, whereas the inner DC-motor speed routine is configured as PI ($K_d = 0$).

TABLE II
FUZZY RULE BASE FOR ONLINE PID GAIN SCHEDULING

$e\backslash\dot{e}$	NB	NS	ZO	PS	PB
NB	NB	NB	NB	NS	ZO
NS	NB	NB	NS	ZO	PS
ZO	NB	NS	ZO	PS	PB
PS	NS	ZO	PS	PB	PB
PB	ZO	PS	PB	PB	PB

The fuzzy supervisor uses attitude error e and a gyroscope-derived rate term $\dot{e}^{(g)}$, with normalization scales of 8° and 4 rad s^{-1}, respectively, mapped to NB, NS, ZO, PS, and PB. Product rule firing, max aggregation, and weighted-average defuzzification use output centers $\{-1, -0.4, 0, 0.4, 1\}$ with the 5×5 rule base in Table II.

To increase near-zero sensitivity, inputs within 30% of their ranges are smoothly amplified by up to 50%, and defuzzified outputs with $|f| < 0.2$ by up to 30%. The gains are $K_p^* = K_p(1 + 1.2f)$, $K_i^* = K_i(1 + 0.08f)$, and $K_d^* = K_d(1 + 0.15f)$. Base gains are $(2, 0.2, 0.02)$ for pitch and $(10, 0.1, 4)$ for roll; the speed loop uses $(1, 0.05, 0)$.

The attitude loops update the integral state as $I_k = \mathrm{clip}(I_{k-1} + e_k \Delta t, -30, 30)$ and the derivative state as $d_k = 0.7 d_{k-1} + 0.3 \dot{e}_k^{(g)}$, where $\dot{e}_k^{(g)} = -\omega_{y,k}$ for pitch and $-\omega_{x,k}$ for roll in the firmware convention. The control output is

$$u_k = K_{p,k}^* e_k + K_{i,k}^* I_k + K_{d,k}^* d_k, \tag{2}$$

A cascaded pitch–speed loop regulates the DC motor, and the roll loop constrains the counterweight to $[-600, 600]$ steps. Output limits, bounded PWM changes, and remote stop commands regulate actuation.

For post-run analysis, water-quality readings are grouped by reference level, and overhead-video positions are used to reconstruct traversal paths. These records are analyzed by test condition; synchronized route-level measurements could support later joint analysis.

IV. EXPERIMENTAL METHOD

Water-quality accuracy was checked separately from locomotion by immersing the packaged probes in labeled paper cups containing pH 4.00, 7.00, and 10.00 standard buffer solutions and turbidity standards of 5, 30, and 80 NTU. Eighteen

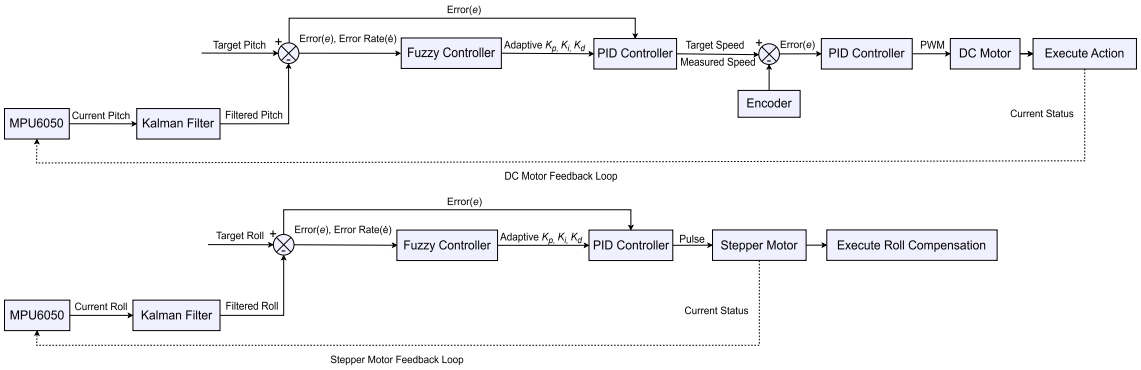


Fig. 4. Feedback-control architecture for the pitch–speed cascade and sliding-counterweight roll loop.

readings were retained at each level and compared with the nominal reference values to characterize bench accuracy and repeatability.

Obstacle-traversal experiments were conducted in a 2 m × 4 m canvas tank filled to 15 cm depth at approximately 25 °C. Each condition was repeated five times. Traversal time was summarized by the mean, whereas maximum trajectory deviation was taken as the largest value observed across the five trials. An overhead camera was used for post-run trajectory measurements.

The robot crossed a 3 m path at a target speed of 0.25 m/s under three conditions: no obstacle, three 5 cm-high stones, and synthetic vegetation covering approximately 30% of the test corridor. Normalized maximum trajectory deviation was estimated from overhead video. Figure 5 shows representative traversal sequences.

For an open-area line-of-sight LoRa field test along the Huangpu River waterfront, the operator remained at the 600-m trail marker while the robot moved from that marker toward the 0-m marker, increasing their separation to approximately 600 m. The Keep route record showed 0.71 km of cumulative travel.

A controlled image-display test presented egret, black-headed gull, and mallard images to the K210 camera, and annotated detection outputs were retained.

Maximum trajectory deviation was normalized by the commanded path length L as

$$d_{\mathrm{rel}} = 100\frac{\max_t |d_\perp(t)|}{L}, \tag{3}$$

where $d_\perp(t)$ is the perpendicular distance between the observed center of the robot and the straight reference path.

V. Results

A. Water-Quality Bench Validation

Across 54 retained pH readings, the pooled MAE was 0.025 pH and the maximum absolute error was 0.05 pH. Across 54 turbidity readings, the maximum relative deviation was 1.00%, occurring at the 5 NTU reference. Table III reports the mean ± sample standard deviation and MAE for each reference level.

TABLE III
Bench Validation of the Water-Quality Module

Metric	Reference	Mean ± SD	MAE
pH	4.00	4.008 ± 0.023	0.019
pH	7.00	7.002 ± 0.029	0.025
pH	10.00	10.004 ± 0.036	0.032
Turbidity	5 NTU	5.016 ± 0.023	0.023 NTU
Turbidity	30 NTU	30.000 ± 0.028	0.022 NTU
Turbidity	80 NTU	80.000 ± 0.032	0.027 NTU

TABLE IV
Traversal Results over a 3 m Path

Condition	Time (s)	Max. dev. (cm)
No obstacle	12.1	3.2
5 cm stones	15.3	7.8
Approx. 30% vegetation	14.7	6.5

B. Mobility, Communication, and Vision

All obstacle-traversal trials reached the endpoint without recorded entrapment. Relative to the unobstructed baseline, mean traversal time increased from 12.1 s to 15.3 s for the stone condition and 14.7 s for synthetic vegetation, corresponding to increases of 26.4% and 21.5%, respectively. The largest recorded maximum trajectory deviation was 7.8 cm under the stone condition, equivalent to 2.6% of the 3 m path (Table IV).

LoRa remote control was maintained as the robot moved away from the operator to approximately 600 m.

Five additional overhead-recorded trials demonstrated terrestrial traversal over a nominal 2.4 m indoor test length. The image-display test retained 191 annotated outputs, including repeated views: 70 egret, 71 black-headed gull, and 50 mallard outputs.

VI. Discussion

The tests show that the implemented prototype can measure water quality, traverse the selected obstacles, and remain remotely controllable at the tested distance. Internal center-of-mass steering avoids an exposed steering mechanism, and the lateral-displacement comparison supported selecting the 10 mm blades. Local filtering reduces short-term noise and

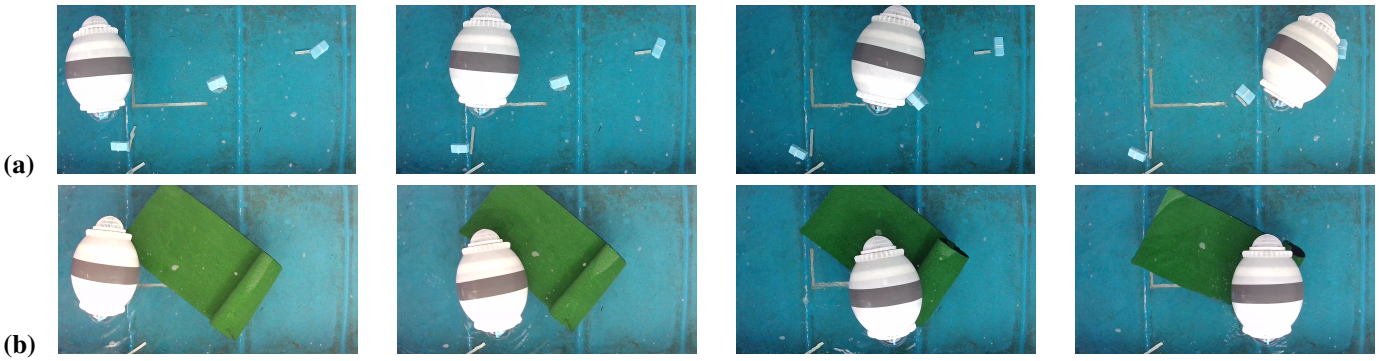


Fig. 5. Time-lapse sequences during obstacle traversal: (a) small-stone condition and (b) synthetic-vegetation condition. Frames are shown in chronological order from left to right.

outlier influence, while modular end rings allow sensor and camera replacement without rebuilding the central drive. This division between the central mechanism and end modules permits a sensing task to be changed while retaining the locomotion platform.

A separate field trial at Xinjiangwan Wetland Park in Shanghai acquired in situ pH and turbidity readings, providing qualitative evidence of field feasibility. The quantitative evaluation uses the controlled bench and tank tests.

The prototype bill of materials was approximately USD 380. The results describe performance under the tested conditions, with condition-level summaries for obstacle traversal and two trajectories per blade configuration. The cup tests characterize the packaged probes at known reference levels, while the tank trials examine mobility in the selected obstacle layouts. The blade comparison identified the 10 mm configuration within its test conditions. Further field evaluation can examine varied terrain, water flow, and ecological disturbance. An integrated field monitoring workflow linking data acquisition, transmission, and environmental analysis remains a direction for further evaluation, while the present vision results are based on image-display tests.

Building on remote control demonstrated at approximately 600 m, future field tests should characterize the maximum usable LoRa range under specified operating conditions and report packet delivery rate (PDR) and received signal strength indicator (RSSI) at representative distances. They could also collect repeated, GPS-tagged water-quality readings along a field route, linking acquisition and transmission to environmental analysis. Such measurements could support Kriging interpolation for spatial water-quality maps, while longer-term records could support analysis of pH and turbidity trends for early warning of wetland degradation. Vision evaluation should use an independent labeled image set and test detection under field conditions. Task-specific models and datasets could extend the camera's applications to floating debris and rare animals.

VII. Conclusion

WetlandSentinel demonstrates an amphibious approach to wetland observation through a streamlined, sealed body and internally driven shell. A three-stage transmission provides propulsion, while a translating counterweight and dual-axis gain-scheduled PID enable directional control and attitude correction. Bench and mobility trials establish operation under the tested conditions, and the field run demonstrates remote control at approximately 600 m. Replaceable sensing assemblies accommodate water-quality probes and onboard visual detection, providing a basis for additional monitoring tasks. Together, these features form an extensible mobile data-acquisition platform for future spatial and temporal environmental data mining.

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